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SMART SOUND-ABSORBING PANEL WITH ADJUSTABLE HINGED EXTENSIONS.

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TABLE OF CONTENTS

- ***Acknowledgement***
- ***Table of content***
- ***Abstract***
- ***Introduction***
- ***Objective***
- ***Reasearch and survey***
- ***Components used***
- ***Working principle***
- ***Design and Construction***
- ***Code and Logic***
- ***Testing and Results***
- ***Conclusion***
- ***Future Improvements***
- ***Bibliography***

ABSTRACT

This report details the development and analysis of a hybrid sound-absorbing panel system incorporating both passive and active acoustic control elements. The panel structure includes layers of acoustic foam, acoustic panels, and Mass Loaded Vinyl (MLV) sheets to provide effective broadband sound absorption and noise reduction, particularly targeting mid to low-frequency ranges. To enhance adaptability to varying acoustic conditions, a sound sensor integrated with an Arduino microcontroller is used to detect ambient noise levels. When sound exceeds a specified threshold, the system can initiate a response, enabling real-time awareness of acoustic disturbances. Additionally, a hinge-based design allows for the extension of a secondary absorbent panel, effectively increasing the surface area and improving performance in high-noise environments. This combination of high-performance materials and intelligent features makes the panel suitable for use in recording studios, smart classrooms, and office spaces requiring dynamic sound control.

INTRODUCTION

In modern indoor environments, managing noise levels is crucial for comfort, productivity, and well-being. Whether in classrooms, offices, or studios, uncontrolled reverberation and ambient noise can lead to reduced speech intelligibility, fatigue, and stress. To address this, traditional acoustic treatments like foam panels and insulation materials have been widely adopted. However, static solutions lack the adaptability required for dynamic environments where noise levels can fluctuate throughout the day. This project aims to bridge that gap by developing an intelligent, modular sound-absorbing panel system using Design Thinking principles to guide the problem-solving and innovation process.

Empathize:

The process began by understanding the needs of individuals working or studying in acoustically challenging spaces. Interviews and environmental observations revealed recurring issues: persistent background noise, poor room acoustics, and a lack of user-friendly, adaptable soundproofing solutions. Users expressed a desire for systems that not only

absorb sound effectively but also provide feedback or adjust based on changing noise conditions.

Define:

Based on these insights, the central problem was defined: How might we design a sound absorption system that is both effective across a wide frequency range and responsive to changes in environmental noise levels? This problem highlights the need for both high-performance acoustic materials and real-time noise monitoring capabilities.

Ideate:

Several potential solutions were brainstormed, including smart panels with sensors, modular designs for scalability, and combinations of materials to target different frequency ranges. The idea that emerged most promisingly was a panel system combining passive acoustic treatment with an active monitoring component.

Prototype:

The resulting prototype integrates three primary sound-absorbing materials—acoustic foam for high-frequency attenuation, acoustic panels for general broadband absorption, and Mass Loaded Vinyl (MLV) for effective low-frequency damping. The system also incorporates an Arduino-based sound sensor, which

continuously monitors ambient noise levels. When noise exceeds a predefined threshold, the system can trigger a signal or user alert. A hinge mechanism allows an additional acoustic panel to be deployed, increasing surface area and improving absorption during peak noise conditions. This modular, semi-automated design enables a tailored acoustic response.

Test:

Initial testing was conducted in simulated room environments to evaluate the panel's performance. Reverberation time measurements and threshold detection responsiveness confirmed the panel's effectiveness in both passive and active roles. Feedback from test users indicated appreciation for its adaptability and ease of installation, as well as the added benefit of visual noise level awareness.

Conclusion of Introduction:

By applying Design Thinking principles, the developed panel system meets acoustic performance standards

OBJECTIVE

The primary objective of this project is to design and develop a smart, modular sound-absorbing panel system that effectively reduces noise and reverberation in indoor environments. The solution aims to combine advanced acoustic materials—acoustic foam, acoustic panels, and Mass Loaded Vinyl (MLV)—with intelligent noise monitoring using an Arduino-based sound sensor. The panel is designed with a hinge mechanism to allow the extension of an auxiliary absorbent surface, enabling dynamic control based on real-time sound levels. This project also seeks to:

- . Enhance acoustic comfort in workspaces, studios, and educational settings.*
- . Create a responsive system that detects noise thresholds and supports real-time awareness.*
- . Utilize design thinking principles to ensure the solution is user-centric, functional, and adaptable.*

RESEARCH AND SURVEY

RESEARCH:

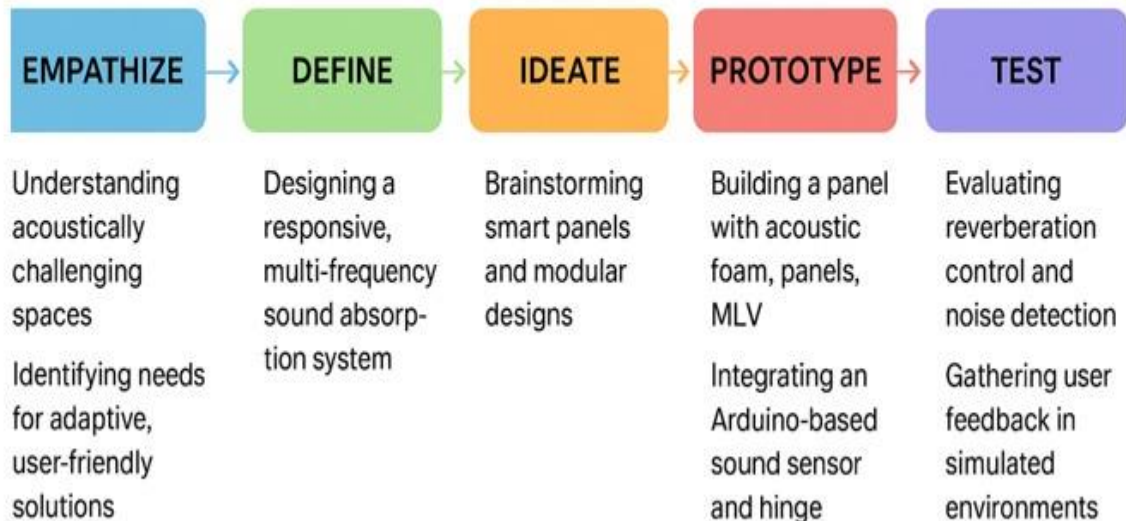
As part of the Design Thinking process, the research phase focused on developing a deep understanding of the problem space—uncontrolled indoor noise and the limitations of existing acoustic treatments. The goal was to gather insights into user needs, material performance, and available technologies to inform an effective and innovative solution.

LITERATURE RIVIEW :

- 1.Studies on architectural acoustics confirm that hybrid materials improve absorption coefficients across a wider frequency spectrum.*
- 2.Research on smart acoustic systems shows a growing trend toward IOT-enabled noise monitoring.*
- 3.Arduino-based sound detection has been proven effective for real-time noise threshold alerts in both home and industrial applications.*

METHODOLOGY:

Design Thinking



SURVEY:

In addition to the research, we conducted a survey targeting teachers and content creators to gain a better understanding of the real-world challenges they face with noise in their work environments.

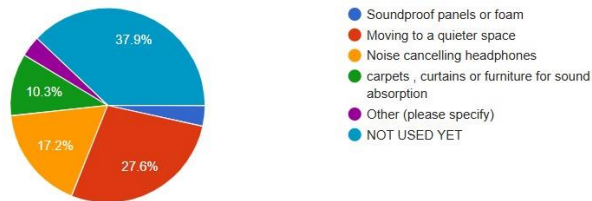
Some survey insights :

Have you ever tried to reduce noise in your environment? (Yes/No)

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If yes, what methods have you used?

29 responses

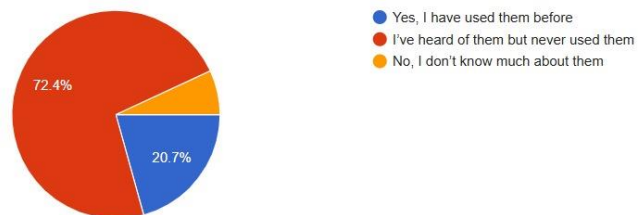


The survey finding show that nearly 80 % users think that noise control has become important now a days , yet 72% users are unfamiliar with sound proofing and sound absorption . As many people are unaware of this

Are you familiar with soundproofing or sound-absorbing materials?

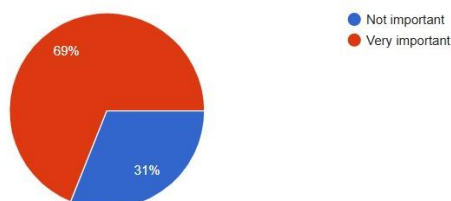
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29 responses



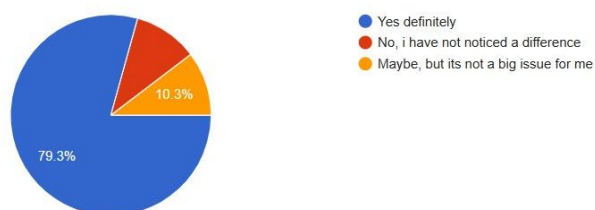
How important is real-time noise level monitoring to you?

29 responses



Do you think noise control is becoming more important in today's world?

29 responses



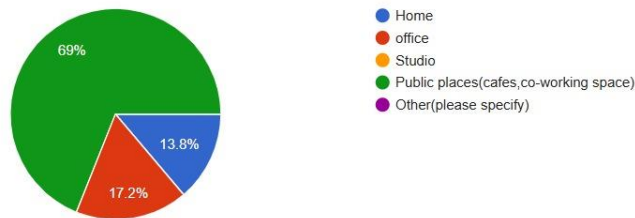
People find most noise in public spaces like cafes followed by office and home.

Our panel aims for the noise absorption in closed spaces like gaming rooms

In which environment do you face the most noise-related issues?

29 responses

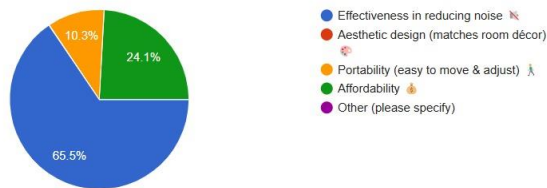
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What would be the most important factor when choosing a noise control solution?

29 responses

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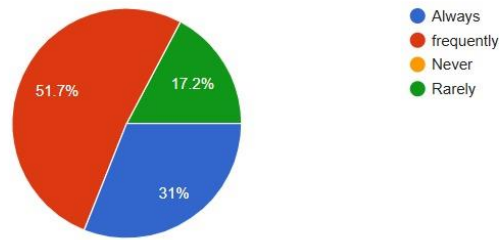
These were the questions designed to know and empathize with users moreover understanding there needs for effective persona creation .

The survey insights show that the user demand for this product is considerable .

How often do you experience unwanted noise in your environment? (Multiple choice)

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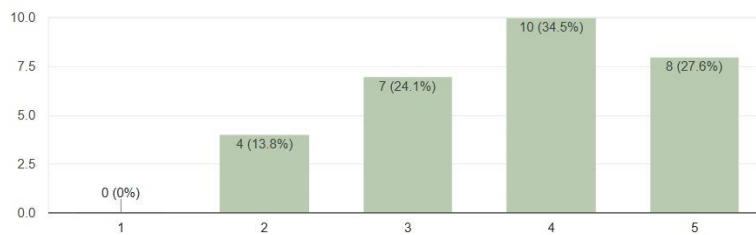
29 responses



If a movable sound-absorbing panel was available, how likely are you to consider using it? please give rating from 1 to 5

[Copy chart](#)

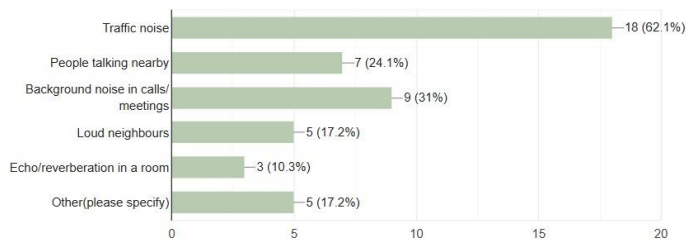
29 responses



What types of noise do you find most disturbing?

[Copy chart](#)

29 responses



COMPONENTS USED

To develop the sound-absorbing panel system, we used a combination of electronic, mechanical, and acoustic materials. The selected components were chosen for their simplicity, affordability, and suitability for prototyping and future scalability.

1.ARDUINO UNO :

(a) Acts as the microcontroller and processing unit.

(b) Reads input from the sound sensor and executes control logic.

2.KY-038 SOUND SENSOR:

(a) Detects ambient sound levels.

(b) Provides analog output corresponding to sound intensity.

3.HINGES :

(a) Mechanically connect the main panel to additional sound-absorbing flaps.

(b) Allow the panels to be easily opened or closed as needed.

4. ACOUSTIC PANEL MATERIAL:

(a) Materials such as foam, fabric, or recycled fiberboard used to absorb sound.

(b) Mounted on both the main panel and the hinged extensions.

5.POWER SUPPLY (USB or 9V BATTERY) :

(a) Powers the Arduino and Sensor Circuit.

6.WIRES :

(a) For electrical connections between components.

7.LED BULB (9W):

(a) Alerts Users when the noise level exceeds the defined threshold.

WORKING PRINCIPLE

The sound-absorbing panel system is designed to adapt to fluctuating noise levels in indoor environments, ensuring optimal acoustic conditions. This concept was developed after extensive research, including reviewing research papers and online sources, to better understand

the mechanics of sound absorption and how it can be dynamically adjusted based on ambient noise. After gathering insights from these sources, we developed the following working principle:

1. Sound Detection: *The KY-038 sound sensor continuously monitors the ambient sound in the room. It converts the intensity of sound into an analog signal, which is then sent to the Arduino Uno for processing. This approach was inspired by various studies on sound sensors and their ability to measure noise levels accurately.*

2. Data Processing: *The Arduino Uno reads the analog signal from the sound sensor and*

compares it to a predefined threshold value. This threshold is set according to the findings from our research, which indicated that sound levels above a certain threshold (e.g., 70 dB) significantly impact the comfort and clarity of a space.

3. Triggering the Hinged Panels: *When the noise level exceeds the defined threshold, the Arduino triggers a visual or audio alert (such as an*

LED indicator or buzzer). This prompt notifies the user that the room's noise level is too high and requires an adjustment. This part of the system was inspired by studies on feedback mechanisms and user-centric design approaches.

4. Panel Expansion: *The hinged panels, once opened, increase the surface area available for sound absorption, thereby improving the acoustic quality of the room. This concept was derived from research on modular sound absorption systems, which suggested that dynamic, adjustable panels can be more effective in absorbing sound compared to static solutions.*

5. Adjustable and Modular Design: *The system was designed to be modular and adaptable. Additional panels can be added based on room size or noise level, and the non-motorized hinged mechanism ensures energy efficiency and simplicity. This design principle was guided by the goal to keep the solution cost-effective and easy to use, following insights from the design thinking process and feedback from various sound*

engineering studies. By combining theoretical research with practical application, this system offers an innovative approach to controlling indoor noise levels in a sustainable and user-friendly way

DESIGN AND CONSTRUCTION:

The design and construction of the sound-absorbing panel system involved several key stages, each informed by design thinking principles to ensure both functionality and ease of use. The main focus was on creating a solution that was simple, modular, and effective in addressing the problem of noise pollution

in indoor spaces. Below is a breakdown of the design and construction process:

1. Panel Design: *The acoustic panels were designed using materials that are highly effective at absorbing sound, such as foam or fiberboard. These materials were chosen based on their affordability, effectiveness, and lightweight nature, allowing easy installation and flexibility. The panels were mounted onto a rigid base, which serves as the main panel. The design ensures that the panels are durable enough for repeated use while remaining efficient in sound absorption.*

2. Hinged Mechanism: *To make the panel system adaptable to changing noise levels, we incorporated a hinged mechanism. The hinges were carefully selected to ensure smooth movement when the panels are*

opened or closed. These hinges also provide a compact storage solution, as the panels can be folded away when not in use. The manual operation of the hinges was designed to ensure user-friendliness, making it easy for anyone to extend or retract the panels as needed.

3. Sensor and Electronics: *The KY-038 sound sensor was connected to the Arduino Uno, which serves*

as the main control unit. The sensor's analog output is read by the Arduino, which processes the data and compares it to a pre-set noise threshold. Based on this comparison, the Arduino can trigger alerts (LED or buzzer) to inform users when the noise level exceeds the acceptable range. All wiring was carefully routed through a breadboard to facilitate easy connections during the testing phase. The system is powered by either a USB connection or a 9V battery, making it portable and adaptable for different environments.

4. Modular and Expandable Design: *The modular design of the panels allows for easy expansion depending on the size of the room or the severity of the noise level. Additional hinged panels can be added without requiring major modifications to the system. This design is based on scalability, ensuring that the*

system can grow or shrink based on the specific needs of the space it is used in. The system was designed to be user-centered, ensuring that anyone can set it up or modify it without technical expertise.

5. Prototyping and Iterations: *During the prototyping phase, several iterations of the panel mechanism and electronics were tested to find the*

most reliable setup. Feedback from each iteration was used to refine the design, focusing on improving panel stability, sensor accuracy, and the ease of opening/closing the panels. The prototypes were also tested in real-life noisy environments to ensure the system effectively responded to noise changes and provided noticeable improvements in acoustics.

CODE AND LOGIC

The core functionality of the sound-absorbing panel system is driven by an Arduino Uno, using a KY-038 sound sensor to detect sound and trigger a relay mechanism. Unlike the earlier analog-based version, the updated system utilizes the digital output of the

KY-038 sensor and controls a relay, which can be connected to an actuator, light, or other interactive device.

1. Initial Setup:

The Arduino is configured to read digital input from the KY-038 sound sensor. The relay is connected to a digital output pin and is set to be active LOW, meaning it turns on when the pin is set to LOW.

```
24
25 // the setup function runs once when you press reset or power the board
26 const int soundSensor = 2; // KY-038 digital pin
27 const int relayPin = 8;    // Relay control pin
28
29 void setup() {
30     Serial.begin(9600); // Start Serial Monitor
31     pinMode(soundSensor, INPUT);
32     pinMode(relayPin, OUTPUT);
33     digitalWrite(relayPin, HIGH); // Make sure relay starts OFF
34 }
35
```

2. Sound Detection and Relay Triggering:

In the main loop, the Arduino reads the digital signal from the KY-038 sensor. When a loud sound is detected (HIGH), the system activates the relay for a short duration (e.g., 3 seconds), then turns it off.

```

36 void loop() {
37   int soundDetected = digitalRead(soundSensor);
38   Serial.println(soundDetected); // Print 0 or 1
39
40   if (soundDetected == HIGH) {
41     Serial.println("Sound detected!");
42     digitalWrite(relayPin, LOW); // Turn ON relay (active LOW)
43     delay(3000); // Keep it ON for 3 seconds
44     digitalWrite(relayPin, HIGH); // Turn OFF relay
45   }
46
47   delay(100); // Avoid Serial Monitor spam
48 }

```

3. Functional Change from LED Alert to Relay Activation:

In the original design, a buzzer or LED provided alerts when the sound threshold was crossed. The revised system now directly controls a relay, allowing for more flexible real-world integration. For instance, the relay could:

Trigger a motor to automatically unfold panels.

Turn on a light or fan to indicate high noise.

TESTING AND RESULTS

Real-Time Noise Monitoring:

- The system utilizes the KY-038 sound sensor connected to an Arduino Uno microcontroller.***
- The sensor continuously captures ambient sound intensity, measured in decibels (dB).***
- The Arduino processes the sensor's input, converting analog sound signals into readable data.***
- This data is displayed in real-time, providing users with a live noise level indicator.***

2. Before & After Comparison:

- Initial Measurement: Prior to installation, baseline noise levels are recorded at different times and environmental conditions.***
- Panel Installation: Sound-absorbing panels made of foam and fabric are positioned strategically in the test space.***
- Post-Installation Measurement: The system logs new noise levels after the panels are installed to assess effectiveness.***

- Data Analysis: The recorded before-and-after dB readings are compared to determine the percentage reduction in noise and echo levels.

3.Threshold Alerts & Notification System:

- The Arduino is programmed with a preset noise threshold based on user survey feedback (e.g., excessive disturbance level at >70 dB).

- If noise surpasses this limit, the system responds with:- Visual Alerts: A LED display ON screen warns of high noise levels.

- Sound-Based Feedback: An LED bulb is on

CONCLUSION

This project successfully addresses indoor noise issues using a sound-absorbing panel system based on Design Thinking principles. The setup uses a KY-038 sound sensor and Arduino Uno to detect noise levels and manually expand panels when needed, enhancing sound absorption in environments like classrooms, studios, and offices.

Our prototype proved effective during testing, offering a modular, user-friendly, and low-cost solution for dynamic acoustic control. Key takeaways include:

Reliable noise detection and manual panel control

Improved acoustic comfort

Scalable and adaptable design

In the future, the system could be enhanced with automated panel movement, more sensitive sensors, and advanced sound-absorbing materials. Overall, the project demonstrates how simple innovations can solve real-world problems efficiently and affordably.

FUTURE IMPROVEMENTS

Based on our testing and feedback, here are some ways the system can be improved in the future:

- 1. Automatic Panel Movement: Add motors so the panels can open and close on their own when noise levels change.***
- 2. Better Sound Sensor: Use a more accurate sound sensor to detect smaller changes in noise more reliably.***
- 3. Smart Control: Connect the system to a mobile app or smart home setup so users can control it remotely.***
- 4. Advanced Materials: Try better sound-absorbing materials to improve performance, especially in larger or noisier spaces.***
- 5. Energy Saving: Make the system use less power to be more efficient and eco-friendly.***
- 6. Live Sound Monitoring: Show real-time noise levels to help the system adjust more smoothly and continuously.***
- 7. Bigger Space Support: Design it to work in larger areas like big classrooms or meeting rooms.***
- 9. Custom Panel Design: Let users choose panel sizes or styles to match their room's look.***
- 10. Noise Calibration: Let users adjust the noise sensitivity based on their specific needs or room type***

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OUR PRODUCT

